Day 039 | Unit 03 | Teacher plan

Integers and Rational Numbers

Compare Order and Distance from Zero

Standards: 6.NS.C.7c, 6.NS.C.7d, 6.NS.C.7a, 6.NS.C.7b

Learning target: I can compare Order and Distance from Zero, and explain and check my reasoning.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. The absolute value |x| is the distance from x to zero, so it cannot be negative. A larger absolute value means farther from zero, but does not necessarily mean a greater number. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 24

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01 / Learn

The absolute value |x| is the distance from x to zero, so it cannot be negative. A larger absolute value means farther from zero, but does not necessarily mean a greater number. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions.

Worked example 1

Find |-10|. Explain the distance meaning.

1. The absolute value |x| is the distance from x to zero, so it cannot be negative.

2. A larger absolute value means farther from zero, but does not necessarily mean a greater number.

3. -10 is 10 units from 0

4. distance is nonnegative.

5. State the result in the form requested: 10.

Answer: 10

Worked example 2

Temperatures are -9 °C and -7 °C. Select all true comparisons.
A. -9 is warmer than -7.
B. -9 < -7
C. -7 lies to the right of -9 on a number line.
D. -9 > 0

1. A number farther right is greater.

2. Among negatives, the number closer to zero is greater.

3. Thus -7 is warmer and is farther right.

Answer: B, C - -9 < -7; -7 lies to the right of -9 on a number line.

Day 039 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find |-40|. Explain the distance meaning.

02 / Constructed response

Compare -5 ___ -4 using <, >, or =. Sketch a number line to justify.

03 / Constructed response

Find |-9|. Explain the distance meaning.

04 / Constructed response

Compare -3/5 ___ -6/5 using <, >, or =. Sketch a number line to justify.

05 / Constructed response

Find |-9/2| and compare it with |3/2|.

06 / Constructed response

Compare -1 ___ -4/3 using <, >, or =. Sketch a number line to justify.

07 / Constructed response

A diver is at -15 m and a bird is at 5/2 m relative to sea level. Who is farther from sea level? Who has greater elevation? Use absolute values.

08 / Multiple select

Temperatures are -6 °C and -4 °C. Select all true comparisons.
A. -6 is warmer than -4.
B. -6 < -4
C. -4 lies to the right of -6 on a number line.
D. -6 > 0

09 / Multiple select

Consider -6 and 4. Select all true statements.
A. -6 is farther from zero than 4.
B. |-6| = -6
C. |-6| = 6
D. -6 > 4

10 / Constructed response

Temperatures are -2°C and -2/5°C. Write an inequality comparing them and explain which is warmer.

Day 039 | Unit 03 | Answer key

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1. 40

-40 is 40 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

2. <

-5 < -4; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

3. 9

-9 is 9 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

4. >

-3/5 > -6/5; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

5. 9/2; 9/2 > 3/2

-9/2 is 9/2 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

6. >

-1 > -4/3; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

7. Diver: 15 m from sea level; bird has greater elevation

|-15| = 15 > |5/2| = 5/2, but -15 < 5/2.

6.NS.C.7c, 6.NS.C.7d | Constructed response

8. B, C - -6 < -4; -4 lies to the right of -6 on a number line.

Among negatives, the number closer to zero is greater. Thus -4 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

9. A, C - -6 is farther from zero than 4.; |-6| = 6

Absolute value measures distance: 6 versus 4. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

10. -2 < -2/5; -2/5°C is warmer

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response